

Stage Two: Requirements

Forward Deployed AI Engineering — Capstone Project

The product requirements document you write from your findings. Everything here must trace back to your discovery workbook. A requirement that cannot be traced is a preference, and preferences do not belong in a PRD.

Week One · Days Four and Five

What This Document Is and Why It Matters

A product requirements document states what is being built, for whom, and how anyone will know when it is finished. It is not a design document and it is not a plan. It should be specific enough that two different engineers reading it would build broadly the same thing, and general enough that it does not dictate how they do it.

You will write version one at the end of week one and you will revise it during week two. That revision is compulsory. Write this version with the expectation that parts of it will turn out to be wrong, and record your assumptions clearly so that you can tell which ones failed when they do.

1. Document Control

Field	Value
Version	1.0
Written by	
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2. The Problem in One Paragraph

CloudServe's support team is answering the same questions repeatedly, slowly, without looking at the documentation that already contains the answers. Seven in ten tickets are ones the agents already know the answer to, yet average resolution takes over seven hours because finding and writing the answer is manual every time. Nearly half of the tickets that reach tier two — at four times the cost of a tier-one resolution — could have been answered at tier one if the

agent had been able to find the right page quickly and confidently. Four of the five project brief targets are within reach from the development data: a retrieval-assisted system that correctly handles answerable tickets can bring FCR from 43.8% to above 60%, drive escalation from 56.2% to below 30%, cut repeat contacts from 21.6% to below 10.8%, and deliver auto-responded tickets a first reply in seconds rather than hours. The one target that cannot be reached through operational improvement alone is the 4.0 CSAT goal — FCR tickets themselves average only 3.41, so even perfect first-contact resolution produces a blended score of 3.41, not 4.0; the gap requires addressing satisfaction drivers that routing does not touch. There is also a problem that exists before any system is built: Latin American customers are already receiving service 10.5 percentage points worse than North American customers on FCR, more than double the five-point governance threshold the brief requires a deployed system to maintain. A system that only preserves that gap is not compliant; closing it must be a design requirement, not a tolerance to manage after launch.

3. Who This Is For

User group	What they need from this system	What they currently do instead	How you will know it worked for them
Customers raising tickets	A first response in minutes rather than hours, with a citation to the documentation source so they can verify the answer before acting on it. Transparency about whether the response is automated.	Wait 8–12 hours. Ravi searches the docs himself and finds the answer about half the time; when the agent reply arrives it says the same thing and he has burned two hours.	First reply time below 5 minutes for auto-responded tickets. Repeat contact rate falls from 21.6% — indicating customers can self-serve from the cited source rather than re-opening the ticket.
Tier one support agents (Sofia)	Intent pre-classified. A draft response with the relevant documentation page attached, surfaced before the agent finishes reading the subject line. Non-fluent tickets restated clearly so the agent answers the right question first time.	Sort the queue by age. Search from memory or a personal snippet file. Compose the answer from scratch. Skip the official documentation because keyword search takes longer than writing from memory.	Agent handling time per ticket falls. The manual search step is replaced by reviewing a draft. The non-fluent ticket resolution time gap (currently +14%) narrows.
Tier two and specialist agents (Daniel)	Every escalated ticket arrives with a structured context packet: intent classification, confidence score, retrieved documentation, and a plain statement of what the system was not confident about. Fewer escalations that could have been resolved at tier one.	Re-read the entire original thread to establish context. Currently receives approximately 49.1% of escalations that were answerable from existing documentation.	Answerable escalations fall from 49.1% of all escalations. Context packet is present on 100% of escalated tickets. Time spent re-reading context on arrival falls.

User group	What they need from this system	What they currently do instead	How you will know it worked for them
Head of support (Marcus)	FCR improvement toward 60% and beyond. Escalation rate below 30%. A decision log that can be presented to the autumn compliance review. Guarantee that no automated response is ever sent for security, compliance, or disputed billing tickets.	Reports first-response time to the executive team as a proxy for FCR. Tracks escalation as a headline count without visibility into the composition of what escalates.	FCR measured directly from system logs, not inferred from response time. Escalation rate falls from 56.2% toward 30%. Zero compliance incidents involving automated responses to must-not-auto tickets.

4. Functional Requirements

Priority scale:

- **Must** — the system is not acceptable without it
- **Should** — expected but could be deferred with a stated consequence
- **Could** — would add value if time allows
- **Will not** — considered and deliberately excluded

ID	Requirement	Priority	Discovery evidence	Acceptance criteria
FR-01	The system shall classify intent and urgency for every incoming ticket and produce a calibrated confidence score before any further processing occurs.	Must	Stage 1 §2: 22 distinct intents across dev tickets; routing correctness depends on classification. §4 governance: confidence within 5pp of observed accuracy. Project Brief §5 routing decision tree.	Every ticket leaving the classify step has a non-null intent, urgency, and confidence score. On the validation set, stated confidence is within 5 percentage points of observed classification precision across confidence bands.
FR-02	The system shall route tickets whose intent is <code>security_incident</code> , <code>compliance_request</code> , <code>feature_request</code> , or <code>unclear_request</code> directly to the human queue with no automated draft generated,	Must	Stage 1 §2: these four intents are 100% <code>must_not_auto_respond</code> in the dev dataset (87/500 tickets, 17.4%). §6 problem statement row 4. Project Brief §5: must-not-auto-respond is the first branch of the routing tree.	Zero tickets with a must-not-auto intent receive a generated draft. Routing log records the must-not-auto decision and the triggering intent for every such ticket. Verified against the 87 development tickets and the validation set.

ID	Requirement	Priority	Discovery evidence	Acceptance criteria
	regardless of confidence score.			
FR-03	The system shall retrieve the most relevant documentation passages for every non-blocked ticket using semantic search against the reviewed documents table. Retrieved content must never originate from ticket history.	Must	Stage 1 §1 Ines: "Someone writes 'my deployment keeps dying' and my article is called 'resolving container health check failures.' There is no path between those two phrases in a keyword search." §3 Step 3; §5 risk register row 2.	For tickets labelled <code>answerable_from_docs = TRUE</code> in the validation set, the correct documentation article appears in the top three retrieved results with $\geq 85\%$ precision. Retrieval source is exclusively the <code>documents</code> table. No ticket body content appears in retrieved results.
FR-04	The system shall generate a draft response grounded in the retrieved documentation, with an explicit citation (article title and section identifier) for every factual claim in the draft.	Must	Stage 1 §1 Sofia: "If it hands me a ticket with a draft and the relevant page attached, that would genuinely help." §1 Ines: wants attribution on every answer to distinguish article errors from retrieval errors. Project Brief §7: citation accuracy $\geq 95\%$.	No sentence in a generated draft makes a factual claim not supported by the cited source. Citation accuracy $\geq 95\%$ on validation set evaluation. Hallucination rate (unsupported claims) $\leq 5\%$. Draft is blocked from release if the validation step (FR-09) fails.
FR-05	The system shall attach a structured context packet to every escalated ticket, containing: classified intent, confidence score, retrieved documentation (if any), and a plain-language statement of what the system was not confident about.	Must	Stage 1 §1 Daniel: "If an escalation arrived saying here is what I think it is about, here is the documentation that seemed relevant, and here is specifically what I was not confident about, I would be twice as fast." §3 Step 6.	100% of escalated tickets carry a populated context packet with all four fields. Zero escalations are forwarded as raw thread-only. Context packet schema: <code>ticket_id</code> , <code>intent</code> , <code>confidence_score</code> , <code>retrieved_doc_ids</code> , <code>uncertainty_statement</code> .
FR-06	The system shall present generated drafts to tier-one agents for review before any response is sent to the customer, for all tickets that do not meet the auto-respond threshold. Agents may	Must	Stage 1 §1 Sofia: prefers draft-and-source over full automation — "I want the draft and the relevant page, not an automated send." §4 Marcus: primary fear is a wrong answer sent without review. §5 risk register row 1.	No draft is sent without an agent action (approve / edit-and-send / discard) unless the ticket meets all three auto-respond conditions: non-blocked intent, confidence $\geq$ threshold, and validation guardrail passed (FR-09).

ID	Requirement	Priority	Discovery evidence	Acceptance criteria
	approve, edit, or discard the draft.			Agent action and timestamp are logged for every ticket.
FR-07	The system shall log every automated routing decision with the following fields: ticket_id, predicted_intent, confidence_score, retrieved_doc_ids, routing_action (auto_respond / draft_for_review / escalate / block), routing_reason, and timestamp.	Must	Stage 1 §4 governance conditions: complete decision logging. Project Brief §4 cross-cutting concerns: "every automated decision must be reconstructable months later." §8 governance obligations item 2.	Decision log contains a row for 100% of processed tickets. A randomly selected logged decision can be fully reconstructed from the log fields alone, without reference to the ticket body. Log is queryable by date, intent, routing_action, and confidence band. Retention minimum 12 months.
FR-08	The system shall normalise incoming tickets from all four channels (email, chat, docs_comment, forum) into a single internal representation, preserving channel as a metadata field available to all downstream components.	Must	Stage 1 §2: four channels with distinct volume and outcome profiles; docs_comment (15.6% of volume) is invisible to most stakeholders and is the highest-value auto-response target. Project Brief §1 channel descriptions.	Tickets from all four channels parse without error. Normalised representation includes: channel, subject, body, received_at, customer_id, customer_tier, customer_region, language_fluency. Channel-specific quirks (forum threading, API comment context) do not corrupt downstream classification or retrieval.
FR-09	The system shall validate every generated draft for private customer data (names, IDs, region, tier) before it reaches the agent queue or is auto-sent, and shall block any draft containing PII sourced from ticket content or ticket history.	Must	Stage 1 §5: PII present in customer_name, customer_id, customer_region fields. §5 risk register row 2. Project Brief §7 governance: zero private data in outbound responses — "there is no acceptable rate."	Zero released responses contain customer PII. Validation step executes on 100% of drafts before release. Tested against a synthetic set of prompts with known PII injection vectors. Blocked drafts are logged and escalated with context packet (FR-05).
FR-10	The system shall produce performance metrics segmented by customer_region,	Must	Stage 1 §1 blind spots: Latin America FCR 38.3% vs North America 48.8% — 10.5pp gap already exceeding the	Fairness report is produced after each evaluation run, showing FCR, escalation rate, and CSAT by region,

ID	Requirement	Priority	Discovery evidence	Acceptance criteria
	customer_tier, and language_fluency — in addition to aggregate metrics — so that quality gaps between groups are visible in every evaluation report.		5pp governance threshold before any system is built. §4 governance conditions. Project Brief §7: quality across customer groups under 5pp difference.	tier, and language fluency. An alert is raised automatically if any pair of groups diverges by more than 5pp on FCR. Report is not optional and is not produced separately from accuracy metrics.
FR-11	The system shall restate the core question from non-fluent English tickets as a normalised English question before classification and retrieval, and shall include the restatement in the agent draft view.	Should	Stage 1 §2: non-fluent tickets take 14% longer to resolve (+58 min avg). §3 Step 2: intent extraction for non-fluent tickets is the highest-value per-ticket automation target. Deferring this increases the handling-time gap but does not block core routing.	Non-fluent ticket restatement is present in the agent draft view. Resolution time gap between fluent and non-fluent tickets narrows from the current +14% baseline on post-launch measurement. Restatement is flagged as AI-generated for agent review.
FR-12	The system shall order the agent review queue by a composite score of SLA time remaining and intent urgency, not by arrival time alone.	Should	Stage 1 §3 Step 1: Sofia sorts by age only; a deployment_failure two hours old may be more urgent than a billing_query six hours old. §1 blind spots: morning cliff (07:00–08:00 UTC tickets average 573–591 min resolution). Deferring this leaves the morning queue-start problem unaddressed.	Queue ordering uses both SLA time remaining and intent urgency tier. A deployment_failure or security_incident with <30 minutes SLA remaining ranks above all lower-urgency tickets regardless of arrival time. Agents can override ordering with a logged reason.

5. Non-Functional Requirements

ID	Category	Requirement	How it will be verified
NFR-01	Latency	End-to-end pipeline latency from ticket receipt to draft delivered to the agent queue must be under 3 seconds at the 95th percentile.	Load test at 50 concurrent tickets. Measure p95 latency from receipt to draft availability. Target applies to auto-respond and draft-for-review paths; the escalate-only path (no generation) is exempt.

ID	Category	Requirement	How it will be verified
NFR-02	Availability	System must be available 99.5% of the time during CloudServe business hours.	Uptime monitoring via Prometheus. Alert on more than 30 minutes cumulative downtime in any business-day window. Monthly availability figure produced from logs.
NFR-03	Classification accuracy	Intent classification precision must reach 85% or better on the validation set. Confidence scores must be calibrated within 5 percentage points of observed precision across confidence bands.	Single evaluation run against the 80-ticket validation set, with date and run count reported. Calibration measured as mean absolute error between stated confidence and observed precision per confidence band. A system that passes 85% aggregate precision but fails calibration does not satisfy this requirement.
NFR-04	Privacy	Zero occurrences of customer PII in any outbound automated response or agent-visible draft sourced from ticket history.	Automated PII scan on 100% of generated drafts before release (FR-09). Tested against a synthetic set of prompts designed to elicit customer name, ID, and region leakage. Zero failures acceptable in any production or staging run.
NFR-05	Auditability	Every automated routing decision must be reconstructable from the decision log alone, without reference to the original ticket body, for a minimum of 12 months after the decision was taken.	Spot-check: an evaluator unfamiliar with the system must be able to reconstruct what happened and why from a decision log entry alone. Log retention configured to 12 months minimum at schema design time, not retrofitted.
NFR-06	Fairness	No two customer groups — defined by region, tier, or language fluency — may differ by more than 5 percentage points on FCR rate among auto-responded tickets after system launch.	Fairness report run after each evaluation cycle. The North America vs Latin America FCR gap (currently 10.5pp, must close to <5pp) is the primary monitored pair. Aggregate accuracy alone does not satisfy this requirement.
NFR-07	Cost	System must operate within the free or near-free tier of the specified stack with no recurring infrastructure spend above a nominal threshold agreed before build.	Cost tracking via provider dashboards throughout the build. No paid-tier service provisioned without explicit approval. Architecture uses local Chroma and all-MiniLM-L6-v2 embeddings to avoid per-query vector store costs.

6. What Is Deliberately Out of Scope

Not building	Why not	What would have to change for this to be reconsidered
Auto-response to security_incident , compliance_request , feature_request , or unclear_request tickets	These four intents are 100% must_not_auto_respond in the development dataset. Security and compliance require human judgment and potentially legal sign-off; feature requests have no retrievable answer; unclear requests require a clarification round-trip before any answer is possible.	A revised intent taxonomy that sub-divides these categories into automatable and non-automatable subtypes, validated by Marcus, Daniel, and legal counsel before any build work.
CSAT improvement to 4.0 / 5	The mathematical ceiling from routing and retrieval improvement alone is 3.41 (the average CSAT of FCR tickets in the development set). Reaching 4.0 requires addressing satisfaction drivers outside the pipeline — documentation quality, product reliability, onboarding, or pricing.	A separate discovery cycle investigating the root causes of FCR ticket CSAT scores below 4.0, leading to a distinct initiative.
Customer-facing chatbot interface	Discovery showed the measured gaps are agent-side (retrieval speed, escalation context) not customer-side (a chat interface). A customer-facing interface adds a new attack surface and a new user group without addressing the 49.1% answerable-escalation problem or the 8-hour response time.	Agent-side system deployed and stable; a second discovery cycle with customers to validate a chatbot hypothesis with data rather than assumption.
Ingestion of agent personal snippet files into the retrieval corpus	These files are unreviewed, contain answers that were correct years ago and may no longer be, and Ines has unsuccessfully requested consolidation twice. Ingesting them risks scaling existing errors.	A governed editorial project led by Ines that reviews, corrects, and formally approves each snippet, producing entries in the documents table under her sign-off.
First-reply SLA timestamp tracking	The development dataset contains only full resolution time, not first-reply timestamp. Building SLA tracking requires a separate capture layer in the production ticketing system — out of scope for this project.	Production integration with the ticketing system that captures and stores the first-reply event timestamp.
Novel incident and roadmap question resolution	No documentation article exists for these categories. The retrieval system cannot surface what has not been written. Any attempt to answer them would generate unsupported content.	Ines writes reviewed articles covering common novel incident patterns; product management maintains a public roadmap FAQ that is

Not building	Why not	What would have to change for this to be reconsidered
		ingested into the documents table.

7. Assumptions and Their Consequences

Assumption	Why you believe it	What happens if it is false	How you will fix it
The 29 reviewed documentation articles are accurate and current within their stated review cycle	Ines confirmed article accuracy in the interview. last_reviewed_days_ago is tracked per article.	The system retrieves accurate-seeming but outdated information. Customers act on stale answers — worse than no answer for compliance-adjacent topics where data_residency already escalates at 69%.	Track last_reviewed_date at retrieval time. Block response for article staleness threshold agent correction rate proxy for stale retrieval
Intent classification can reach 85%+ precision across all 22 intent categories on approximately 500 training examples	Average of ~22 examples per intent in the development set. This is a low-data regime; precision will vary widely by category.	The routing decision rests on unreliable input. Must-not-auto tickets may be misclassified and receive drafts; high-volume answerable intents may be misrouted.	Evaluate per-intent on the validation set deployment. Any cases below 80% precision mandatory human review regardless of aggregate accuracy.

Assumption	Why you believe it	What happens if it is false	How you will fix it
The 10.5pp FCR governance gap between North America and Latin America is closeable by improving retrieval quality equally across regions, without operational or staffing changes	The Latin America gap persists even among fluent customers (CSAT 2.64 vs non-fluent 3.00), suggesting a retrieval or routing cause rather than a language cause.	The gap has a structural driver (staffing timezone, contract tier differences, routing priority) that retrieval improvement cannot address.	Segment post-launch by region and language fluency. If the gap persists six weeks after deployment, escalate to an operational investigation with M
The <code>ground_truth_responses.json</code> reference answers are representative of what a good tier-one response looks like	Written by a senior support agent; covers the answerable subset of the development tickets. They are the only structured evaluation reference available.	Evaluation against ground truth optimises for the wrong target; scores look good while customer experience does not improve.	Compare ground truth evaluation scores against live agent satisfaction on the same sample. If the system is in production, Large divergence suggests ground truth quality to be corrected before Stage 5 evaluation.
The confidence threshold can be calibrated using the development and validation sets to a level that satisfies both the 85% precision requirement and the governance calibration condition	Threshold calibration is standard practice; the project brief acknowledges 0.80 as illustrative, not fixed.	The threshold is set by intuition rather than measurement. Over-confident routing sends drafts that should escalate; over-cautious routing escalates everything and saves no agent time.	Calibration curve or validation set. If measured absolute error between stated confidence and observed precision is 5pp, the threshold-setting approach must be reviewed before Stage 5 sign-off
The four-channel ingest can be normalised to a common representation without losing information material to routing or retrieval	The development dataset already contains a normalised representation across all four channels, suggesting this is feasible.	Channel-specific context (e.g. the forum thread that preceded a comment, the API documentation page a <code>docs_comment</code> is attached to) is lost. Routing quality	Per-channel classification precision in the evaluation material gap between email/chat and <code>docs_comment</code> /forum precision signals the channel-specific context is being lost.

Assumption	Why you believe it	What happens if it is false	How you will fix it
		degrades for those channels.	

8. Success Measures

Baselines use development data actuals (n=500) as the truth anchor. Project Brief figures differ on three measures and are shown for reference. CSAT 4.0 is not achievable through routing improvement alone — the FCR ceiling is 3.41 (see Stage 1 §4 achievability analysis). All baselines are pre-system; post-launch figures require production data capture.

Measure	Baseline (dev data)	Brief baseline	Target	Measured how	Reported by
First contact resolution	43.8% (219/500)	42%	60% or better	first_contact_resolution rate on post-launch tickets, combining auto-responded and agent-reviewed outcomes	Head of support, from system dashboard
Time to first reply	Not in dataset (resolution avg: 422 min)	8–12 hours	Under 5 minutes for auto-responded tickets	Interval from ticket received_at to first outbound response, captured by the ingest layer in production	System monitoring dashboard
Customer satisfaction	2.97 / 5	3.2 / 5	3.2 as first milestone; 3.41 is the operational ceiling — reaching beyond it requires a separate initiative	Mean CSAT rating on tickets where the system was involved, segmented by routing path	Head of support, same reporting mechanism as today
Escalation rate	56.2% (281/500)	58%	30% or lower	Tickets escalated to tier two divided by total tickets processed by the system	Head of support, from system dashboard

Measure	Baseline (dev data)	Brief baseline	Target	Measured how	Reported by
Repeat contacts	21.6% (108/500)	Not previously measured	10.8% or lower (halved, per brief)	Tickets from the same customer on the same issue within 7 days of the previous interaction	System dashboard; requires customer_id linkage across tickets in production
Regional FCR gap	10.5pp (NA 48.8% vs LatAm 38.3%)	Not previously measured	Under 5pp between any two regions	FCR segmented by customer_region, produced in fairness report alongside every evaluation cycle	Fairness report — mandatory, not optional

9. Open Questions

Question	Why it matters	Owner	Resolve by
What is the exact date of the autumn compliance review Marcus mentioned?	The decision log schema and explainability design must be in place before that review. If it falls within the project timeline it changes the build order.	Marcus Adeyemi	Before Stage 3 prompt library is finalised
Does the production ticketing system capture a first-reply timestamp distinct from resolution time?	The development dataset contains only resolution time. First-reply SLA tracking — the metric Marcus actually presents to the executive team — requires a separate timestamp. Without it, the primary success measure for SLA compliance cannot be verified.	Marcus Adeyemi / ticketing system owner	Before Stage 4 sprint plan
What is the correct auto-respond confidence threshold, and who has authority to move it post-launch?	The 0.80 figure in the brief is illustrative. Too low sends drafts the system cannot defend; too high escalates everything and saves no agent time. Threshold ownership and change control must be agreed before launch.	Evaluator (capstone) + Marcus Adeyemi	After classifier calibration in Stage 5
Which specific sub-questions within data_residency drive the 69% escalation rate,	data_residency is the top-volume intent and the highest-effort category (515 min avg). If the gap is a documentation coverage problem, Ines needs to write new articles	Ines Varga + Daniel Okonkwo	Before Stage 5 system evaluation

Question	Why it matters	Owner	Resolve by
and are they covered by existing documentation?	before the retrieval system can help. If it is a contract-specific question that documentation cannot answer, it should be added to the must-not-auto list.		
Is the communication channel between Sofia and Ines functional enough to support a feedback loop from agent corrections to the knowledge base?	Sofia and Ines disagree on whether the search problem was communicated. If that channel is broken, a feedback path from agent-flagged wrong answers back to the knowledge base will also fail — and the knowledge base will drift from what the system retrieves.	Sofia Restrepo + Ines Varga	Before Stage 4 sprint plan

The review before you move on: Read your finished PRD and ask, for each requirement in turn, whether an engineer who had not attended any of your interviews could build it from what is written. Where the answer is no, the requirement is not finished. This check takes twenty minutes and it is the single highest-value thing you can do at the end of week one.

Related documents: [Stage_1_Discovery_Workbook.md](#) · [Stage_3_Prompt_Library.md](#) · [Stage_5_PRD_Revision_Log.md](#)